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meets the criteria for a verbal warning, but also for a written reprimand, and hence the fifth option is not a key.

The correct answer is *Absent 4.5 hours on 19 March 2010; stayed one hour late on 20 March 2010; absent all day on 8 February 2011 and 9 February 2011; arrived 40 minutes late on 17 April 2011.*

***Written reprimand*:**

As noted in the analysis for *Verbal warning*, the employee in the fifth option meets the criteria for both a verbal warning and a written reprimand, and hence this option is the answer. Also by the previous analysis, the employees in the first, second, and fourth options do not meet the criteria for verbal warning, and hence cannot meet the criteria for written reprimand. While a verbal warning is warranted in the third option, the employee has fewer than 4 absences in a 365-day period and thus does not merit a written reprimand.

The correct answer is *Absent 3 hours on 7 July 2010; absent 2.5 hours on 13 September 2010; absent all day on 31 January 2011 and 4 July 2011; absent 5 hours on 12 March 2011.*

•

Marketing strategist: Agency A designed an advertising campaign that our company is about to test with a focus group. We are wondering whether a new ad campaign will increase our name recognition among consumers. As a contingency, we have decided that we might ask Agency B to design an alternate campaign. However, if we find that Agency A's campaign elicits positive responses from the focus group, we will not ask Agency B for a campaign.

A statement that must be true if the marketing strategist's statements are true: After the focus testing is complete, if the company ___1___, then it must also be the case that the company ___2___.

Select for *1 if* and *2 then* the two different options that create a statement that must be true if the marketing strategist's statements are true. Make only two selections, one in each column.

1 if	2 then	
		believes Agency A's campaign will increase the company's name recognition
		believes Agency A's campaign will decrease the company's name recognition
		asks Agency B for a campaign
		asks Agency A for an alternate campaign
		believes the focus group did not respond positively to Agency A's campaign

Answer Explanation:

Infer

The item requires completing a statement such that if a certain claim about a company is true, then a different claim about the company must also be true. The passage indicates that the company will not ask Agency B to design an advertising campaign if Agency A's advertising campaign elicits positive responses from the focus group. Therefore, if the company *asks Agency B for a campaign*, then it must be the case, given the information provided, that the company *believes the focus group did not respond positively to Agency A's campaign*. There are no other relationships between any two answer options that would create a statement that follows logically from the information provided.

***1 if*:**

The correct answer is *asks Agency B for a campaign.*

***2 then*:**

The correct answer is *believes the focus group did not respond positively to Agency A's campaign.*

•

Francois and Pierre each owe Claudine money. Today, Francois will make a payment equal to 50% of the amount he owes Claudine, and Pierre will make a payment equal to 10% of the amount he owes Claudine. Together, the two payments will be equal to 40% of the combined amount that Francois and Pierre owe Claudine.

Select for *Francois* and *Pierre* amounts that Francois and Pierre could owe Claudine that are jointly consistent with the given information. Make only two selections, one in each column.

Francois	Pierre	

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		€50
		€250
		€750
		€3,750
		€6,750

Answer Explanation:

Infer

***Francois*:**

Let F and P be the amounts, in euros (€) and respectively, that Francois and Pierre owe Claudine. Then $0.4(F+P)$ is 40% of the total amount they owe Claudine, and $0.5F+0.1P$ is the total amount they will pay Claudine today.

$$\begin{aligned} 0.5F+0.1P &= 0.4(F+P) && \text{given} \\ 5F+P &= 4(F+P) && \text{multiply both sides by 10} \\ 5F+P &= 4F+4P && \text{expand right side} \\ F &= 3P && \text{subtract } 4F+P \text{ from both sides} \end{aligned}$$

The table below shows the value of F (which is $3P$) for each possible value of P.

$F=3P$	P
150	50
750	250
2,250	750
11,250	3,750
20,250	6,750

The only value of $F=3P$ above that is one of the allowable values is 750.

The correct answer is *€750*.

***Pierre*:**

The table above shows that 250 is the value of P corresponding to $F=750$.

The correct answer is *€250*.

•

Witness testimony: Around 8:00 on the morning of Tuesday, July 6, I saw a man exit from the back entrance of the building. I knew it was McGregor by the man's height and his long, slightly awkward steps, and from the fact that he was wearing a hat—possibly a baseball cap. McGregor always seemed to be wearing some kind of hat. On Tuesdays there are always several groundskeepers scheduled to work on the grounds, and he would very likely have been scheduled that day.

Select *Assumes to be true* for the statement that describes something the testimony implies the witness assumes to be true, and select *Assumes to be false* for the statement that describes something the testimony implies the witness assumes to be false. Make only two selections, one in each column.

Assumes to be true	**Assumes to be false**	
		McGregor was the only groundskeeper scheduled to work at the building on Tuesday, July 6.
		McGregor exited the rear of the building in question alone at 8:00 on the morning of Tuesday, July 6.
		McGregor was scheduled to work at the building on the morning of Tuesday, July 6.

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		McGregor's height, way of walking, and tendency to wear hats distinguish him from the other groundskeepers.
		McGregor often wore a baseball cap prior to the morning of Tuesday, July 6.

Answer Explanation:

Infer

***Assumes to be true*:**

The witness says there are always several groundskeepers scheduled to work on the grounds on Tuesdays. But the witness also claims to have known that the man seen exiting the building on the Tuesday morning was McGregor, based on the man's height, way of walking, and wearing of a hat. Since the witness assumes that these features revealed that it was McGregor rather than another groundskeeper who was seen exiting the building, the witness is also assuming that McGregor's height, way of walking, and tendency to wear hats distinguish him from the other groundskeepers.

The correct answer is *McGregor's height, way of walking, and tendency to wear hats distinguish him from the other groundskeepers.*

***Assumes to be false*:**

The witness claims there are always several groundskeepers scheduled to work on the grounds on Tuesdays. This claim implies that more than one groundskeeper was scheduled to work there on Tuesday, July 6. So, the witness is assuming it's false that McGregor was the only groundskeeper scheduled to work there that day.

The correct answer is *McGregor was the only groundskeeper scheduled to work at the building on Tuesday, July 6.*

•

Ribonucleic acid (RNA) is a molecule built from sequences of smaller molecules called nucleobases. RNA nucleobases are of 4 different types: adenine (A), cytosine (C), guanine (G), and uracil (U). Consider the collection of all possible RNA sequences consisting of 12 nucleobases, 3 of each type. An RNA sequence will be selected at random from this collection and the first 3 nucleobases of the sequence will be detached from the sequence.

In the table, select the probability that the 3 nucleobases are all of the same type, and select the probability that they are of 3 different types. Make only two selections, one in each column.

All the same type	3 different types
	$\frac{1}{55}$
	$\frac{9}{220}$
	$\frac{9}{110}$
	$\frac{9}{55}$
	$\frac{27}{55}$

Answer Explanation:

Apply

***All the same type*:**

We're considering a randomly ordered sequence of 12 nucleobases, which is made up of 3 of each of the 4 types of nucleobase. The first nucleobase in this sequence must be of the same type as exactly 2 of the remaining 11 nucleobases. So, the probability that the first 2 nucleobases are of the same type is $\frac{2}{11}$. If they are of the same type, then exactly 1 of the remaining 10 nucleobases is also of that type. Thus, the probability that the first 3 nucleobases in the sequence are all of the same type is $\frac{2}{\ } \times \frac{1}{\ } = \frac{1}{\ }$

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the sequence are all of the same type is $\frac{}{11} \times \frac{}{10} = \frac{}{55}$.

The correct answer is $\frac{1}{55}$.

***3 different types*:**

We saw above that the first nucleobase in the sequence is of the same type as exactly 2 of the remaining 11 nucleobases. So, it's of a different type than each of the other 9 of those 11. That means the probability that the first 2 nucleobases are of different types is $\frac{9}{11}$. If the first

2 nucleobases are of different types, then among the 10 remaining nucleobases are still 3 apiece of the other 2 types, making a total of 6 nucleobases that are of types different than either of the first 2 nucleobases' types. In that case, the probability that the third nucleobase in the sequence is of a different type than either of the first 2 is $\frac{6}{10}$.

Thus, the probability that the first 3 nucleobases in the sequence are all of different types is $\frac{9}{11} \times \frac{6}{10} = \frac{27}{55}$.

The correct answer is $\frac{27}{55}$.

•

Let S be the set of all 7-digit numbers N such that the sum of the digits of N is divisible by 7.

A and B are two 7-digit numbers:

$A = 4{,}893, \mathrm{P}12$

$B = 5{,}857{,}0\mathrm{Q}5$

Select a value for P so that A is an element of S. Select a value for Q so that B is an element of S. Make two selections, one in each column.

P	Q	
		4
		5
		6
		7
		8

Answer Explanation:

Apply

P**:**

The sum of the digits in A other than P is 4 + 8 + 9 + 3 + 1 + 2 = 27. Since 8 + 27 = 35, which is divisible by 7, selecting the value 8 for P makes A a 7-digit number whose digits sum to a value divisible by 7. And that makes A an element of S.

The correct answer is *8*.

Q**:**

The sum of the digits in B other than Q is 5 + 8 + 5 + 7 + 0 + 5 = 30. Since 5 + 30 = 35, which is divisible by 7, selecting the value 5 for Q makes A a 7-digit number whose digits sum to a value divisible by 7. And that makes B an element of S.

The correct answer is *5*.

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